

FitBuddy AI — Data Flow & Technology Stack

Phase 2 • Requirement Analysis

Source note: Technology details are taken from the supplied FitBuddy project documentation.

High-level data flow

- Profile form → POST /api/users → SQLite profile persistence → POST /api/users/{user_id}/plans/generate → plan generation path → plan displayed in UI.
- User feedback → POST /api/users/{user_id}/feedback → feedback stored → rule-based adjustment → next Generate again → refined plan.
- Coach/API endpoints expose summaries and operational data.

Technology stack

Layer	Technology	Purpose
Backend	Python + FastAPI	Web server and REST API
Server	Uvicorn	Local application server
Validation	Pydantic	Request/response validation
Configuration	python-dotenv + .env	Environment configuration
Database	SQLite	Local persistence
AI	Gemini integration (optional)	AI-assisted plan generation
Frontend	HTML, CSS, JavaScript	User and coach interfaces
Testing	pytest	Automated tests